

FORT VERMILION, AB, Canada

WMO: 710240

Lat: 58.3822N

Lon: 116.0403W

Elev: 289

StdP: 97.90

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.3	-32.4	-39.2	0.1	-35.2	-35.8	0.1	-32.3	6.5	-7.7	5.5	-12.2	0.8	300	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	28.9	18.0	27.2	17.1	25.5	16.2	19.7	26.1	18.6	24.8	17.6	23.3	2.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	13.0	21.7	16.4	12.1	20.6	15.4	11.3	19.7	57.4	25.9	54.0	24.9	50.8	23.5	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.8	4.3	-39.9	32.4	3.9	1.7	-42.7	33.6	-44.9	34.6	-47.1	35.5	-49.9	36.8		
(5)			-39.6	21.3	3.8	1.8	-42.3	22.6	-44.5	23.6	-46.6	24.6	-49.4	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-17.9	-15.3	-8.2	2.9	11.1	16.0	18.2	15.9	10.1	2.1	-9.7	-17.2		
	DBStd	14.42	7.83	7.30	8.18	5.79	5.03	3.43	2.88	3.47	4.33	4.74	6.72	7.41		
	HDD10.0	4140	866	708	564	219	47	3	0	2	50	248	591	843		
	HDD18.3	6483	1124	941	822	462	229	84	38	91	247	502	841	1102		
	CDD10.0	764	0	0	0	7	79	183	253	184	54	4	0	0		
	CDD18.3	65	0	0	0	0	3	13	33	14	1	0	0	0		
	CDH23.3	929	0	0	0	4	86	206	408	198	26	1	0	0		
	CDH26.7	199	0	0	0	0	14	36	106	41	3	0	0	0		
	Wind	WSAvg	1.8	1.6	1.7	2.0	2.3	2.3	1.8	1.6	1.8	2.0	1.7	1.5		
(15) Precipitation	PrecAvg	409	22	20	23	21	37	49	68	58	37	29	22	22		
	PrecMax	573	64	39	59	71	97	132	177	115	97	83	38	45		
	PrecMin	295	6	2	3	0	6	8	18	14	10	3	0	10		
	PrecStd	82	12	11	15	18	23	34	38	30	22	21	12	10		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.3	5.1	11.8	22.4	28.7	30.0	31.9	30.2	26.5	20.1	7.5	2.4		
		MCWB	1.3	1.7	5.3	11.1	14.6	18.0	19.7	18.9	17.2	12.3	3.5	0.3		
	2%	DB	-1.4	1.3	8.5	17.4	25.7	27.5	29.3	27.5	22.8	14.7	4.0	-1.8		
		MCWB	-2.3	-0.9	3.2	8.6	13.1	16.2	18.8	18.0	14.5	8.8	1.6	-2.6		
	5%	DB	-4.4	-2.3	6.2	14.6	23.4	25.8	27.5	25.5	20.2	11.8	1.3	-4.5		
		MCWB	-4.9	-3.8	1.8	6.8	12.1	15.7	18.3	16.9	13.2	7.1	0.1	-4.8		
	10%	DB	-7.4	-5.0	3.6	12.2	20.8	23.9	25.6	23.6	17.8	9.2	-0.9	-6.9		
		MCWB	-7.7	-6.0	0.3	5.6	10.9	15.1	17.3	15.9	11.9	5.8	-1.5	-7.2		
	0.4%	WB	1.2	2.1	5.4	11.4	15.6	19.8	21.8	20.7	18.0	12.9	3.9	0.8		
		MCDB	3.0	5.1	11.2	21.5	26.2	25.9	28.1	27.4	25.2	19.4	6.4	1.9		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-2.2	-1.1	3.6	9.0	14.2	18.3	20.0	19.2	15.8	9.5	1.8	-2.5		
		MCDB	-1.5	1.0	8.0	16.7	23.7	24.4	26.9	24.7	21.0	13.3	3.6	-1.7		
	5%	WB	-4.9	-3.8	2.0	7.2	12.7	17.0	19.1	18.0	14.2	7.4	0.1	-4.8		
		MCDB	-4.3	-2.4	5.7	13.5	21.4	22.9	25.7	23.6	18.4	11.1	1.1	-4.5		
	10%	WB	-7.7	-6.0	0.4	5.9	11.4	16.0	18.2	16.9	12.8	6.0	-1.6	-7.3		
		MCDB	-7.4	-5.0	3.4	11.6	19.4	21.7	24.1	22.1	16.6	8.8	-0.9	-6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	11.0	12.6	12.2	13.3	12.3	12.4	12.4	11.8	8.5	7.4	7.9		
		MCDBR	12.1	14.1	14.8	15.9	17.4	15.8	15.9	15.8	15.7	13.6	9.2	9.4		
		MCWBR	11.1	11.8	10.1	8.5	7.4	6.6	6.9	7.1	8.0	8.1	7.3	8.7		
	5% WB	MCDBR	12.0	13.9	14.3	14.9	15.4	13.1	14.3	14.3	14.1	12.7	8.8	9.3		
		MCWBR	11.0	11.7	9.8	8.3	7.0	6.0	6.8	7.1	7.6	7.8	7.3	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.222	0.246	0.272	0.331	0.342	0.361	0.369	0.348	0.311	0.283	0.233	0.203		
	taud		2.235	2.313	2.301	2.333	2.426	2.401	2.389	2.447	2.546	2.559	2.339	2.122		
	Ebn at Noon		658	817	886	878	892	875	858	854	841	772	645	543		
	Edh at Noon		43	66	90	105	103	107	107	94	72	54	39	34		
(44) All-Sky Solar Radiation	RadAvg		0.57	1.51	3.18	4.88	5.71	5.96	5.64	4.57	3.09	1.52	0.60	0.34		
	RadStd		0.04	0.10	0.20	0.23	0.43	0.45	0.26	0.29	0.22	0.18	0.07	0.03		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	+1.59	+2.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page